

Solutions – Practice Problem Set 1

ECON 308 – International Economic Relations

Fall 2026

1. **Gravity intuition.** Larger GDPs imply more production and income, scaling the number of potential buyers and sellers; microfoundations include love-of-variety models and market-size effects that raise the mass of matches. Distance proxies trade costs (freight, time-to-market, information frictions), which depress the intensive and extensive margins of trade. Search/information models also imply distance lowers match probabilities, while per-unit transport costs directly reduce delivered profitability.
2. **Relative wages under free trade.** With perfect competition and specialization, each country's wage equals the value of marginal product in the sector it produces: $w = \frac{P_g}{a_{Lg}}$. The *relative* wage across countries is pinned down by world prices and comparative advantage: the high-wage country must have sufficiently high productivity in its export good to remain competitive at world prices.
3. **Unit labor costs and prices.** In autarky, competitive pricing implies $P_g = w a_{Lg}$, so relative prices equal unit labor cost ratios. With trade, world prices need not equal a country's own unit costs; instead, specialization ensures the country produces the good where it has the lowest unit cost at world prices, and wages adjust to equalize unit costs in the produced sector.

4. Gravity Model: Baseline Predictions and Elasticities.

Given: $T_{ij} = 10^{-6} \frac{Y_i Y_j}{D_{ij}}$, with $(Y_A, Y_B, Y_C) = (2000, 1000, 500)$ (billions), and distances in km: $D_{AB} = 6,000$, $D_{AC} = 1,500$, $D_{BC} = 2,000$.

(a) *Predicted bilateral exports.*

$$\begin{aligned} T_{AB} &= 10^{-6} \frac{(2000)(1000)}{6000} = 10^{-6} \cdot \frac{2,000,000}{6000} = 10^{-6} \cdot 333.\bar{3} = 0.000333 \text{ (billions)} \\ &\Rightarrow \$0.333 \text{ million.} \end{aligned}$$

$$\begin{aligned} T_{AC} &= 10^{-6} \frac{(2000)(500)}{1500} = 10^{-6} \cdot \frac{1,000,000}{1500} = 10^{-6} \cdot 666.\bar{6} = 0.000667 \text{ (billions)} \\ &\Rightarrow \$0.667 \text{ million.} \end{aligned}$$

$$\begin{aligned} T_{BC} &= 10^{-6} \frac{(1000)(500)}{2000} = 10^{-6} \cdot \frac{500,000}{2000} = 10^{-6} \cdot 250 = 0.000250 \text{ (billions)} \\ &\Rightarrow \$0.250 \text{ million.} \end{aligned}$$

- (b) *Effect of a 10% fall in distance on T_{AB} .* The log-specification gives the elasticity of trade with respect to distance as $-\gamma = -1$. Exactly:

$$\frac{T_{AB}^{\text{new}}}{T_{AB}^{\text{old}}} = \left(\frac{D_{AB}^{\text{new}}}{D_{AB}^{\text{old}}} \right)^{-\gamma} = (0.9)^{-1} = \frac{1}{0.9} = 1.\bar{1}$$

So T_{AB} rises by 11.1%.

- (c) *Effect of 5% GDP increases in both A and B.* Because $T_{ij} \propto Y_i Y_j$,

$$\frac{T_{AB}^{\text{new}}}{T_{AB}^{\text{old}}} = (1.05)(1.05) = 1.1025 \Rightarrow \text{increase of 10.25\%}.$$

- (d) *Log residual against observed \$1.3 bn.*

$$\ln T_{AB}^{\text{obs}} - \ln T_{AB}^{\text{pred}} = \ln(1.3) - \ln(0.000333 \dots) \approx 8.269.$$

A positive residual indicates AB trade exceeds gravity prediction, consistent with omitted frictions/facilitators (e.g., common language, trade agreement, high-quality transport links, or substantial services trade not captured by simple distance).

5. Two-Country, Two-Good Ricardian Basics.

Given: Unit labor requirements

	a_{LW}	a_{LC}	
H	2	4	$L_H = L_F = 240.$
F	6	3	

- (a) *Opportunity costs and comparative advantage.* In the Ricardian model,

$$\text{OC}_H(W \text{ in } C) = \frac{a_{LW}}{a_{LC}} = \frac{2}{4} = 0.5, \quad \text{OC}_H(C \text{ in } W) = \frac{a_{LC}}{a_{LW}} = \frac{4}{2} = 2.$$

$$\text{OC}_F(W \text{ in } C) = \frac{6}{3} = 2, \quad \text{OC}_F(C \text{ in } W) = \frac{3}{6} = 0.5.$$

Home (H) has lower OC in W ($0.5 < 2$), so H has comparative advantage in W ; Foreign (F) in C .

- (b) *PPFs and autarky prices.* PPFs satisfy $a_{LW}W + a_{LC}C \leq L$.

$$H: 2W + 4C \leq 240 \Rightarrow W = 120 - 2C \text{ (slope } -2). \quad F: 6W + 3C \leq 240 \Rightarrow W = 40 - \frac{1}{2}C \text{ (slope } -\frac{1}{2}).$$

Under perfect competition, autarky relative price equals the unit-cost ratio:

$$\left(\frac{P_W}{P_C} \right)_H^A = \frac{a_{LW}}{a_{LC}} = \frac{1}{2} = 0.5, \quad \left(\frac{P_W}{P_C} \right)_F^A = \frac{6}{3} = 2.$$

- (c) *Range for complete specialization.* With H in W and F in C ,

$$0.5 < \frac{P_W}{P_C} < 2.$$

(d) *Specialization at $P_W/P_C = 1.2$.* Since $0.5 < 1.2 < 2$, H specializes in W , F in C .

$$H \text{ output: } W_H = \frac{L_H}{a_{LW}} = \frac{240}{2} = 120, \quad C_H = 0. \quad F \text{ output: } C_F = \frac{L_F}{a_{LC}} = \frac{240}{3} = 80, \quad W_F = 0.$$

H exports W and imports C ; F the reverse.

(e) *Real wages under free trade.* With specialization, $w_H = \frac{P_W}{a_{LW}}$ and $w_F = \frac{P_C}{a_{LC}}$.

$$H: \frac{w_H}{P_W} = \frac{1}{a_{LW}} = \frac{1}{2} = 0.5 \text{ (units of } W), \quad \frac{w_H}{P_C} = \frac{P_W/P_C}{a_{LW}} = \frac{1.2}{2} = 0.6 \text{ (units of } C).$$

$$F: \frac{w_F}{P_C} = \frac{1}{a_{LC}} = \frac{1}{3} \text{ (units of } C), \quad \frac{w_F}{P_W} = \frac{1}{a_{LC}} \cdot \frac{P_C}{P_W} = \frac{1}{3 \cdot 1.2} = \frac{1}{3.6} \approx 0.278 \text{ (units of } W).$$

6. Terms of Trade and Real Incomes (price falls to $P_W/P_C = 0.8$).

(a) Since $0.5 < 0.8 < 2$, specialization pattern is unchanged: $H \rightarrow W$, $F \rightarrow C$.

(b) *Real wages at $P_W/P_C = 0.8$.*

$$H: \frac{w_H}{P_W} = \frac{1}{2} = 0.5 \text{ W}, \quad \frac{w_H}{P_C} = \frac{0.8}{2} = 0.4 \text{ C}. \quad F: \frac{w_F}{P_C} = \frac{1}{3} \text{ C}, \quad \frac{w_F}{P_W} = \frac{1}{3 \cdot 0.8} = \frac{1}{2.4} \approx 0.417 \text{ W}.$$

Comparison to autarky: Autarky real wages are $H : (1/2 \text{ W}, 1/4 \text{ C})$; $F : (1/6 \text{ W}, 1/3 \text{ C})$. Both countries gain from trade relative to autarky.

(c) *Terms of trade (TOT) effects.* H exports W , so its TOT is P_W/P_C . When this falls from 1.2 to 0.8, H 's purchasing power in C falls ($0.6 \rightarrow 0.4$), while F 's TOT $= (P_C/P_W)$ improves (its real wage in W rises $0.278 \rightarrow 0.417$).

7. World Relative Supply with Three Countries.

Given:

	a_{LW}	a_{LC}	
1	1	2	$L_1 = L_2 = L_3 = 100.$
2	2	3	
3	4	5	

Compute a_{LW}/a_{LC} : Country 1: 0.5; 2: $2/3 \approx 0.667$; 3: $4/5 = 0.8$.

(a) *Ordering by comparative advantage in W :* 1 (best), 2, 3.

(b) *World relative supply (WRS).* Each country produces W if $P_W/P_C \geq a_{LW}/a_{LC}$; otherwise C .

Price range	W producers	W supply	C supply
$P < 0.5$	none	0	$\frac{100}{2} + \frac{100}{3} + \frac{100}{5} = 50 + 33.\bar{3} + 20 = 103.\bar{3}$
$0.5 < P < 2/3$	1	$\frac{100}{1} = 100$	$\frac{100}{3} + \frac{100}{5} = 33.\bar{3} + 20 = 53.\bar{3}$
$2/3 < P < 0.8$	1, 2	$\frac{100}{1} + \frac{100}{2} = 150$	$\frac{100}{5} = 20$
$P > 0.8$	1, 2, 3	$\frac{100}{1} + \frac{100}{2} + \frac{100}{4} = 175$	0

Thus the step values of $RS^{W/C} \equiv W/C$ are: 0; $\frac{100}{53.\bar{3}} \approx 1.875$; $\frac{150}{20} = 7.5$; and $+\infty$. Kinks at $P = 0.5, 2/3, 0.8$.

- (c) *Equilibrium with $RD^{W/C} = 1.1 - 0.2(P_W/P_C)$.* Because RD is continuous and decreasing, and RS is a step function, the market clears either (i) on a step (if RD hits a step value), or (ii) at a kink with the marginal country partially specialized. Evaluate RD at kinks:

$$RD(0.5) = 1.1 - 0.2(0.5) = 1.0, \quad RD(2/3) \approx 0.967, \quad RD(0.8) = 0.94.$$

All are *below* the adjacent step value 1.875. Hence equilibrium occurs at the *lowest* kink $P_W/P_C = 0.5$, with Country 1 splitting labor between W and C to make $RS = RD = 1$. Let $x \in [0, 1]$ be the share of Country 1's labor in W at $P = 0.5$.

$$W = \frac{100x}{1} = 100x, \quad C = \frac{100(1-x)}{2} + \frac{100}{3} + \frac{100}{5} = 50(1-x) + 53.\bar{3} = 103.\bar{3} - 50x.$$

Set $W/C = 1$:

$$100x = 103.\bar{3} - 50x \Rightarrow 150x = 103.\bar{3} \Rightarrow x \approx 0.6889.$$

Production at equilibrium: Country 1 uses $\approx 68.9\%$ of L_1 in W ; Countries 2 and 3 produce only C .